Chapter 17 Homework Answer Sheet

Name:________ Student ID:________ Instructor:________ Score:________

Part A multiple choice (15 points)

1-5: EBDAC

6-10: AABDC

11-15: EBDED

Part B short question （15 points）

16. (3 points)

(**a**) Stay the same.
(**b**) The concentration of B(*aq*) increases.
(**c**) pH of the solution decreases.

17. (6 points)

(**a**)

$$\mathrm{pH} = \mathrm{p}K_\mathrm{a} + \log\frac{[\mathrm{base}]}{[\mathrm{acid}]}$$

$$[\mathrm{base}] = [\mathrm{sodium\ acetate}] = \frac{0.13\ \mathrm{mol}}{1.00\ \mathrm{L}} = 0.13\ M$$

$$[\mathrm{acid}] = [\mathrm{acetic\ acid}] = \frac{0.10\ \mathrm{mol}}{1.00\ \mathrm{L}} = 0.10\ M$$

$$\mathrm{pH} = -\log(1.8\times10^{-5}) + \log\frac{0.13}{0.10} = 4.7447 + 0.11394 = 4.8587 = 4.86$$

(**b**)

$$[\mathrm{base}] = [\mathrm{sodium\ acetate}] = \frac{0.13+0.02}{1.00\mathrm{L}} = 0.15\ M$$

$$[\mathrm{acid}] = [\mathrm{acetic\ acid}] = \frac{0.10-0.02}{1.00} = 0.08\ M$$

	$CH_3COOH(aq)$	⇌	$CH_3COO^-(aq)$	+	$H^+(l)$
In. Con.	0.08 M		0.15 M		0
Ch. Con.	-x M		(0.15+x) M		x M
Eq. Con.	(0.08-x) M		(0.15+x) M		x M

$$K_\mathrm{a} = \frac{x(0.15+x)}{0.08-x} = 1.8\times10^{-5}$$

$$[\mathrm{H}^+] = x \approx 9.6\times10^{-6}$$

$$\mathrm{pH} = -\log[\mathrm{H}^+] = 5.0177 = 5.02$$

OR

$$\mathrm{pH} = -\log(1.8\times10^{-5}) + \log\frac{0.15}{0.08} = 4.7447 + 0.2730 = 5.0177 = 5.02$$

(**c**)

$$[\mathrm{base}] = [\mathrm{sodium\ acetate}] = \frac{0.13-0.02}{1.00\mathrm{L}} = 0.11\ M$$

$$[\mathrm{acid}] = [\mathrm{acetic\ acid}] = \frac{0.10+0.02}{1.00} = 0.12\ M$$

	$CH_3COOH(aq)$	⇌	$CH_3COO^-(aq)$	+	$H^+(l)$
In. Con.	0.12 M		0.11 M		0
Ch. Con.	-x M		(0.11+x) M		x M
Eq. Con.	(0.12-x) M		(0.11+x) M		x M

$$K_\mathrm{a} = \frac{x(0.11+x)}{0.12-x} = 1.8\times10^{-5}$$

$$[\mathrm{H}^+] = x \approx 1.964 \times 10^{-5}$$
$$\mathrm{pH} = -\log[\mathrm{H}^+] = 4.7069 = 4.71$$

$$\mathrm{pH} = -\log(1.8 \times 10^{-5}) + \log\frac{0.11}{0.12} = 4.7447 - 0.03779 = 4.70691 = 4.71$$

18. (6 points)

(**a**)

$$K_{sp} = [\mathrm{Ag}^+][\mathrm{I}^-] = 8.3 \times 10^{-17}$$
$$[\mathrm{Ag}^+] = [\mathrm{I}^-] = 9.11 \times 10^{-9} = 9.1 \times 10^{-9}\ M$$

(**b**) calculate the equilibrium constant for the reaction AgI(*s*) + 2 CN^-(*aq*) ⇌ $Ag(CN)_2^-$(*aq*) + I^-(*aq*),

$$K = K_{sp} \times K_f = 8.3 \times 10^{-17} \times 1.0 \times 10^{21} = 8.3 \times 10^4$$

(**c**)

K is much greater than one for the reaction of AgI(*s*) with CN^-. This means that the reaction goes to completion. For a AgI(*s*) in 0.100 *M* NaCN solution, CN^- is the limiting reactant. Two moles of CN^- react with one mole of AgI, so the solublility of AgI in 0.100 *M* NaCN is (0.100/2) = 0.0500 *M.*

OR

AgI(*s*) + 2 CN_2^-(*aq*) ⇌ $Ag(CN_2)^-$(*aq*) + I^-(*aq*) ,
0.100-2x x x

$$K = \frac{[\mathrm{Ag(CN_2)^-}][\mathrm{I}^-]}{[\mathrm{CN_2}^-]^2} = \frac{x^2}{(0.100 - 2x)^2} = 8.3 \times 10^4$$

$$[\mathrm{I}^-] = x = 0.05\ M$$